

```
print(string.hexdigits)
#print(string.whitespace)
print(string.punctuation)
```

Output:

C:/Users/MRCET/AppData/Local/Programs/Python/Python38-32/pyyy/strrmodl.py

```
=====
abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ
0123456789
0123456789abcdefABCDEF
!"#$%&'()*+,-./:;<=>?@[\\]^_`{|}~
```

Python String Module Classes

Python string module contains two classes – Formatter and Template.

Formatter

It behaves exactly same as str.format() function. This class becomes useful if you want to subclass it and define your own format string syntax.

Syntax: from string import Formatter

Template

This class is used to create a string template for simpler string substitutions

Syntax: from string import Template

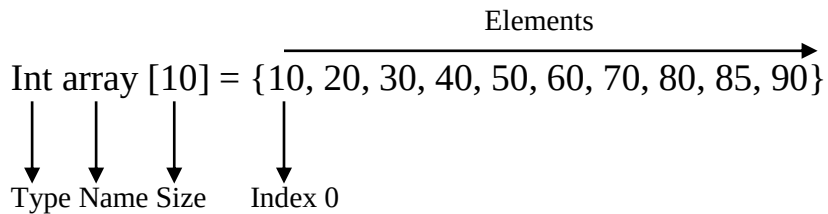
Python arrays:

Array is a container which can hold a fix number of items and these items should be of the same type. Most of the data structures make use of arrays to implement their algorithms. Following are the important terms to understand the concept of Array.

- **Element**– Each item stored in an array is called an element.
- **Index** – Each location of an element in an array has a numerical index, which is used to identify the element.

Array Representation

Arrays can be declared in various ways in different languages. Below is an illustration.



As per the above illustration, following are the important points to be considered.

- Index starts with 0.
- Array length is 10 which means it can store 10 elements.
- Each element can be accessed via its index. For example, we can fetch an element at index 6 as 70

Basic Operations

Following are the basic operations supported by an array.

- Traverse – print all the array elements one by one.
- Insertion – Adds an element at the given index.
- Deletion – Deletes an element at the given index.
- Search – Searches an element using the given index or by the value.
- Update – Updates an element at the given index.

Array is created in Python by importing array module to the python program. Then the array is declared as shown below.

```
from array import *
```

```
arrayName=array(typecode, [initializers])
```

Typecode are the codes that are used to define the type of value the array will hold. Some common typecodes used are:

Typecode	Value
b	Represents signed integer of size 1 byte
B	Represents unsigned integer of size 1 byte

c	Represents character of size 1 byte
i	Represents signed integer of size 2 bytes
l	Represents unsigned integer of size 2 bytes
f	Represents floating point of size 4 bytes
d	Represents floating point of size 8 bytes

Creating an array:

```
from array import *  
array1 = array('i', [10,20,30,40,50])  
for x in array1:  
    print(x)
```

Output:

```
>>>
```

```
RESTART: C:/Users/MRCET/AppData/Local/Programs/Python/Python38-32/arr.py
```

```
10  
20  
30  
40  
50
```

Access the elements of an Array:**Accessing Array Element**

We can access each element of an array using the index of the element.

```
from array import *  
array1 = array('i', [10,20,30,40,50])  
print (array1[0])  
print (array1[2])
```

Output:

RESTART: C:/Users/MRCET/AppData/Local/Programs/Python/Python38-32/pyyy/arr2.py

10

30

Array methods:

Python has a set of built-in methods that you can use on lists/arrays.

Method	Description
<u>append()</u>	Adds an element at the end of the list
<u>clear()</u>	Removes all the elements from the list
<u>copy()</u>	Returns a copy of the list
<u>count()</u>	Returns the number of elements with the specified value
<u>extend()</u>	Add the elements of a list (or any iterable), to the end of the current list
<u>index()</u>	Returns the index of the first element with the specified value
<u>insert()</u>	Adds an element at the specified position
<u>pop()</u>	Removes the element at the specified position
<u>remove()</u>	Removes the first item with the specified value
<u>reverse()</u>	Reverses the order of the list
<u>sort()</u>	Sorts the list

Note: Python does not have built-in support for Arrays, but Python Lists can be used instead.

Example:

```
>>> college=["mrcet","it","cse"]

>>> college.append("autonomous")

>>> college

['mrcet', 'it', 'cse', 'autonomous']

>>> college.append("eee")

>>> college.append("ece")

>>> college

['mrcet', 'it', 'cse', 'autonomous', 'eee', 'ece']

>>> college.pop()

'ece'

>>> college

['mrcet', 'it', 'cse', 'autonomous', 'eee']

>>> college.pop(4)

'eee'

>>> college

['mrcet', 'it', 'cse', 'autonomous']

>>> college.remove("it")

>>> college

['mrcet', 'cse', 'autonomous']
```